

BOSS Engineering Design Input Criteria

Project Name: BIXXX – BOSS Dealer Flash Tool

Project Engineer: Ben Lockwood

Signature / Date: 8/4/2025

Engineering Manager: Gary Blank

Signature / Date: 8/4/2025

Engineering Director: Jon Coyne

Signature / Date: 8/4/2025

Marketing Product Manager: Cory Jokela

Signature / Date:

Date: 5/28/2025

Revision Date: 5/28/2025

Project Description:

The box must be able to flash all spare parts control components identified below, as well as simulate a vehicle for flashing of equipment in the field when a truck is not easily accessible (i.e., assembled equipment in a warehouse).

Design Inputs:

Functional Requirements

1. Interface with BOSS vehicle systems for diagnostics,
 - Firmware Flashing
 - Updating configuration variables with Diag Tool
2. Box must be easily expandable to accommodate future unknown connectors for unreleased control components. This includes but not limited to:
 - Plow Hand Controller
 - Spreader Controller
 - Plow Module
 - Spreader Module
 - MAG Display
 - B.O.B
 - V.I.M
 - Single Stage, Tri Motor, and Trimotor with GPS
 - Brine Maker Controller
3. Portable for easy transportation and setup in both shop and field environments.
4. Designed for minimal training and plug-and-play operations.
5. A toggle switch to flip between using AC/DC and 18V Battery.
6. The box must provide clear indication of what power mode is enabled (i.e. 110v AC or 18v battery, etc)
7. Be able to simulate ON/OFF Key Switch
8. Provide easily replaceable automotive fuse, or explore circuit breakers.
9. Provide termination resistors
 - This is 100% required. The box would not function correctly without them. We will need at least 1 somewhere on the CAN network. CAN

standards states "For low-speed/fault-tolerant CAN, each device on the network needs a termination resistor for each data line"

10. Use of adaptor harness to eliminate wear on the box connections.

Electrical Requirements

1. Explore feasibility of utilizing 110v AC for in-shop use, and a battery or other form of remote power for field use. Must understand all regulatory and compliance requirements and provide a recommended path forward.
2. Box must be powerable by 110v AC (i.e., able to plug into wall)
 - Extension Cord (NEMA 5-15P to NEMA 5-15R)
 - Must provide 12V DC ? @ Amps.
3. 18V Battery

Performance Requirements

4. The device remains operational at ambient temperatures of –20C to 50C

Applicable Standards (Statutory, Regulatory, and Industry, etc) – Final list still TBD

5. Weather/Water Proofing
 - Ingress Protection Rating: IP44(?)
6. UL CSA Standard Requirements
7. Other Standards (TBD)

Cost Requirements

8. Design cost target sub-100\$/Box.

Connection Requirements

9. Box must provide clean (minimize or eliminate any “hanging” harnesses) connection point(s) to all existing and future BOSS control components.
10. Current required connections:
 - Harness #1
 - i. Hand Controller – Plow
 - ii. Display – Spreader
 - iii. VBX Controller – Spreader
 - iv. Plow Controller – Plow

- Harness #2
 - i. MAG display – MX150 20 pin.
 - ii. Tri-Motor – Deutsch 4 pin
 - iii. Bluetooth (BOB)
 - iv. VIM
- Harness #3
 - i. TBD (For future Controllers)
- Box must provide a clean connection to CFT bumper plug with minimal harnessing residing outside of the box. Must be able to connect to the largest AWG wire connection in BOSS product lineup.

Note: Design input criteria should be updated as needed during product development.

PIN Outs for Harnesses and Connectors

Assembly	Connector	CAN High	CAN Low	Key-Switch	Power	Ground
SnowCM	MX150 16 pin.	4	5	1	8	7
Hand control	Micro 10 pin	5	4	3	2	1
Display control	Micro 8 pin	2	3	7	1	4
VBX+ motor	35 pin	23	22	28	20,29,31	18
VIM	TBD					
Tri Motor	Deutsch 4 pin	1	4	2	X	3
Tri Motor	METRI-PACK 2 PIN	X	X	X	B	A
MAG Display	MX150 20 pin	10	9	16	4	13
Bluetooth (BOB)	Deutsch 6 pin	1	6	5	4	3
Adaptor harness	Deutsch 6 pin	1	6	5	4	3
Panel Mount	Deutsch 6 pin	1	6	5	4	3
Panel Mount	Deutsch 3 pin	A	B	X	X	X